qBraid features we used

We built an eavesdropper alarm for a power-grid control link, using the BB84 quantum key protocol.

Quantum hardware

IonQ, IQM, Rigetti, Quantinuum

IQM Garnet and IonQ Forte 1: 2 jobs each, clean link vs. eavesdropper

Simulators

GPU state vector, tensor network

GPU state vector: tuned the alarm line on 30 qubits before touching hardware

qBraid Lab

Notebooks, environments

One shared notebook for the team, with a Launch on qBraid button

Encouraged

qBraid AI

AI Chat, Agent mode, MCP server

An agent drafted our batched circuit, and AI Chat compared device prices

Transpiler

Qiskit, Cirq, Braket, OpenQASM

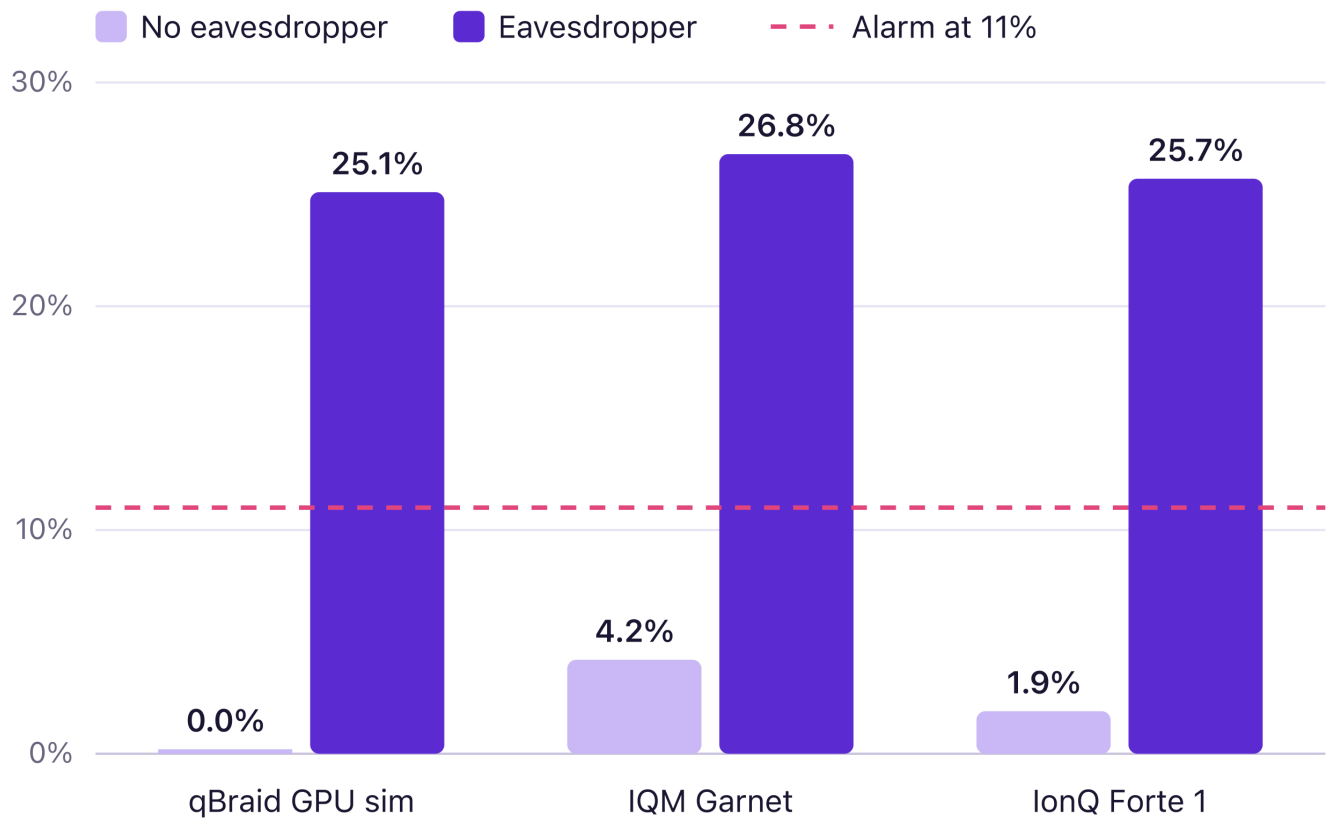
Wrote it once in Qiskit, and qBraid converted it for each device

Jobs and results

Jobs page, job IDs, plots

Pulled job IDs from the Jobs page and plotted all three backends together

Our alarm works on real quantum hardware



Illustrative numbers. Error rate = share of key bits that disagree.

What it shows

Noise alone stays under our 11% alarm line. An eavesdropper pushes every backend past 25%.

Where AI helped

An agent drafted the batched circuit in minutes. It mixed up which bits to keep, and we caught it on the simulator.

What surprised us

IQM Garnet ran noisier than IonQ, but still left a wide margin under 11%.

Check it

Proof

- IQM Garnet: <ID 1>, <ID 2>
- IonQ Forte 1: <ID 3>, <ID 4>
- Job IDs copied from the Jobs page in our qBraid account

Links

- Screen recording (1:45): <link>
- qBraid notebook: <link>
- Repo with a Launch on qBraid button: <link>

Cost and next

- About 1,100 credits (about $11), 4% of our budget
- Batched every key bit into one circuit per run: 4 hardware jobs in total
- Next: run the alarm on Quantinuum

Spending more credits doesn't score. A careful, cheap run beats an expensive one.